Laminar flow

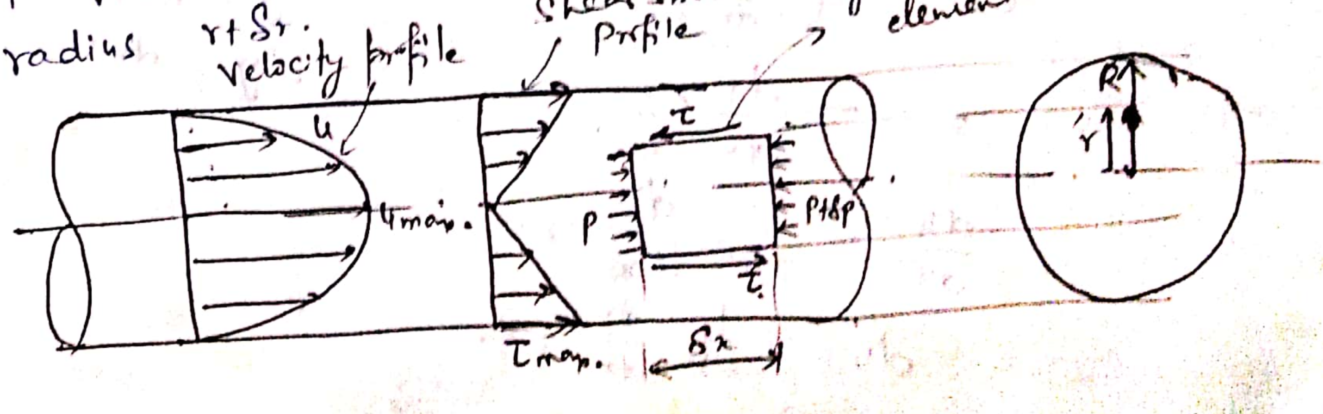
- Fluid particles move along straight parallel paths in layers or laminae.
- Laminar flow occurs at low velocity so that forces due to viscosity predominate over inertial forces.
- Fluid rise over each other giving rise to shear stresses being maximum at boundary. Shear stresses so produced develop resistance to flow.
- In order to overcome resistance pressure drops from point to point along flow direction.
- Relationship between shear and pressure gradient in laminar flow

$$\frac{\partial \tau}{\partial y} = \frac{\partial p}{\partial x}$$

Steady laminar flow through a circular pipe:

Hagen poiseullie's law

Consider steady laminar flow of an incompressible real fluid through a horizontal circular pipe. Since the flow proceeds in circular laminae, we take equilibrium of the lamina of cylinder of radius r contained in the lamina of radius r+δr.



②

For equilibrium net pressure force on cylindrical element due to pressure P & P+δP must be equal to shear force on its periphery due to presence of cylindrical lamina over it.

$$P \times \pi r^2 - (P+\delta P)\pi r^2 = \tau \times 2\pi r \times \delta x$$

$$\tau = -\frac{dP}{dx}\frac{r}{2} \quad \text{---------- (i)}$$

If laminar flow in a pipe is fully developed, The velocity profile do not change in longitudinal direction, the velocity u varies only with r and Pressure P remains constant over the cross section.

Hence $\frac{dP}{dx}$ remains constant.

$\boxed{\frac{dP}{dx} = \text{Constant}} \Rightarrow \boxed{\tau \propto r}$ Shear stress varies linearly with radius.

τ_{max} is at $r = R$

$$\tau_{max} = -\frac{dP}{dx}\frac{R}{2}$$

Velocity distribution

For laminar flow

$$\tau = \mu\frac{du}{dy}$$

→ Since R is taken from centre

→ y is measured from wall

$y = R - r$

$dy = -dr$

$$\tau = \mu\left(-\frac{du}{dr}\right)$$

$$= -\frac{dP}{dx}\frac{r}{2}$$

$$\frac{du}{dr} = \frac{1}{2\mu}\frac{dP}{dx}r$$

$$du = \frac{1}{2\mu}\frac{dP}{dx}r\,dr$$

$$u = \frac{1}{2\mu}\frac{dp}{dx}\frac{r^2}{2} + c$$

B.C. at $\boxed{r = R, \quad u = 0}$

$$\Rightarrow c = -\frac{1}{4\mu}\frac{dp}{dx}(R^2)$$

$$u = \frac{1}{4\mu}\frac{dp}{dx}(r^2 - R^2)$$

$$\boxed{u = -\frac{1}{4\mu}\frac{dp}{dx}(R^2 - r^2)}$$ ← Parabolic velocity distribution.

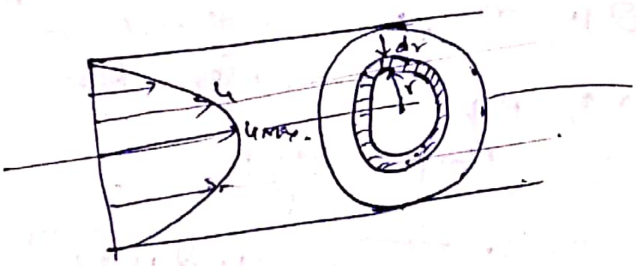
B.C at $r = 0$, $u = u_{max}$.

$$u_{max} = -\frac{1}{4\mu}\frac{dp}{dx}R^2$$

$$\boxed{u = -\frac{1}{4\mu}\frac{dp}{dx}R^2\left[1 - \frac{r^2}{R^2}\right]}$$

$$\boxed{u = u_{max}\left[1 - \frac{r^2}{R^2}\right]}$$

Ratio of max^{m} velocity to average velocity.



Total discharge (Q) passing through pipe can be obtained by integrating small discharge passing through elemental ring of thickness dr.

$$dq = [(2\pi r)\,dr] \times u$$

$$q = \pi R^2 u_{mean}.$$

$$= \int_0^R u(2\pi r\,dr)$$

$$\pi R^2 u_{mean} =$$

$$u_{mean} = \frac{1}{\pi R^2}\int_0^R u\,dA$$

$$\left[\begin{array}{l} A = \pi R^2 \\ dA = 2\pi r\,dr \end{array}\right]$$

$$\boxed{u_{mean} = \frac{1}{A}\int_0^R u\,dA}$$

(v)

$$u_{mean} = \frac{1}{\pi R^2}\int_0^R u_{max.}\left(1-\frac{r^2}{R^2}\right) 2\pi r\,dr$$

$$\boxed{u = u_{max.}\left(1-\frac{r^2}{R^2}\right)}$$

$$u_{mean} = \frac{u_{max.}}{\cancel{\pi} R^2}\, 2\cancel{\pi}\int_0^R \left(r - \frac{r^3}{R^2}\right)dr$$

$$= \frac{2u_{max}}{R^2}\left[\frac{r^2}{2} - \frac{r^4}{4R^2}\right]_0^R = \frac{2u_{max}}{R^2}\cdot\frac{R^2}{4}$$

$$u_{mean} = \frac{u_{max}}{2} \Rightarrow \boxed{u_{max.} = 2u_{mean}}$$

Discharge

$$q = A \times V$$

$$= \frac{\pi}{4} D^2 \times u_{mean.} = \frac{\pi}{4} D^2 \times \frac{u_{max.}}{2}$$

$$= \frac{\pi}{4} D^2 \times -\frac{1}{4\mu}\frac{dp}{dx}\frac{R^2}{2}\ \ D^2/4$$

$$\boxed{q = -\frac{\pi D^4}{128\mu}\frac{dp}{dx}}$$

Hagen poiseullie's equation for laminar flow through a circular pipe.

as $x \uparrow$, $p \downarrow$ therefore -ve.

Note:
$$\frac{dp}{dx} = -\frac{128\mu q}{\pi D^4}$$

$$\frac{P_1 - P_2}{L} = \frac{128\mu q}{\pi D^4} \Rightarrow \boxed{P_1 - P_2 = \frac{128\mu q L}{\pi D^4}}$$

$$\boxed{P_1 - P_2 = \frac{32\mu V L}{D^2}}$$

$$h_f = \frac{P_1 - P_2}{\rho g} = \frac{128\mu q L}{\pi D^4 \cdot \rho g}$$

$$= \frac{128\mu \times \left(\frac{\pi}{4}D^2 \times V\right) \times L}{\pi D^4 \rho g}$$

$$\boxed{h_f = \frac{32\mu V L}{\rho g D^2}}$$

Hagen's poiseullies formular for laminar flow.

$h_f \propto V$
$h_f \propto \frac{1}{D^2}$
for lamina flow.

Power to maintain the flow

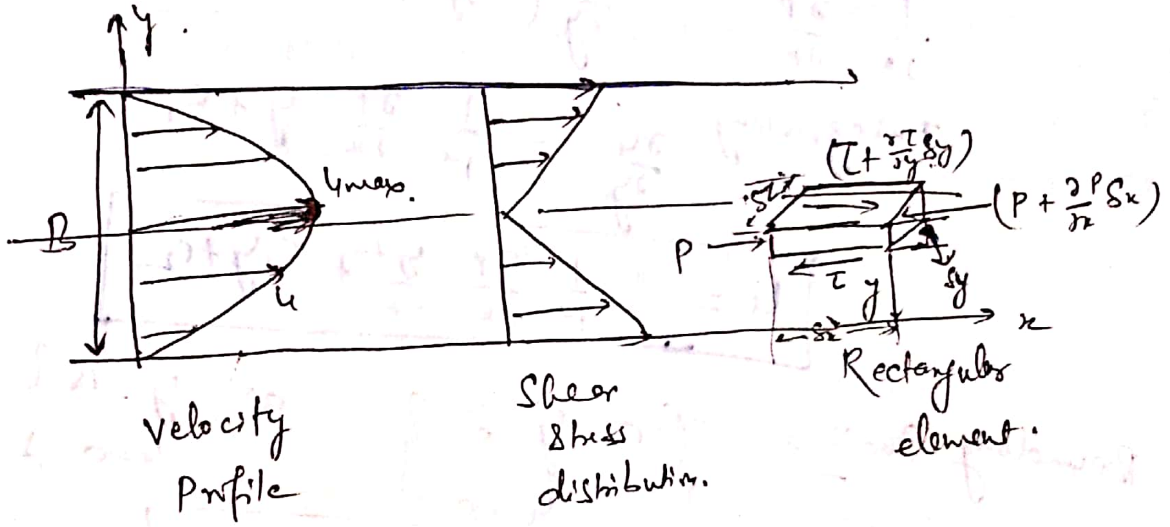
$$\boxed{\text{Power} = \left[(P_1 - P_2) \times \pi R^2\right] \times V}$$

Total drag force on pipe

$$\boxed{D = \tau_w \times 2\pi R \times l}$$

Laminar flow between parallel plates

Both plates at rest: Plane poiseulli's flow



Consider rectangular element at distance y. In order to balance shearing force in fluid, a pressure gradient in the direction of flow must be maintained.

$$\left[P - \left(P + \frac{\partial P}{\partial x}\delta x\right)\right]\delta y\,\delta z = \left[\tau - \left(\tau + \frac{\partial \tau}{\partial y}\delta y\right)\right]\delta z\,\delta x$$

$$-\frac{\partial P}{\partial x}\cancel{\delta x}\,\cancel{\delta y}\,\cancel{\delta z} = -\frac{\partial \tau}{\partial y}\cancel{\delta y}\,\cancel{\delta x}\,\cancel{\delta z}$$

$$\Rightarrow \boxed{\frac{\partial P}{\partial x} = \frac{\partial \tau}{\partial y}}$$

(8)

This means that in steady laminar flow, the pressure gradient in the direction of flow is equal to shear stress gradient in normal direction.

<u>Velocity distribution</u>

$$\tau = \mu \frac{du}{dy} \Rightarrow \frac{\partial p}{\partial x} = \frac{\partial}{\partial y}\left(\mu \frac{du}{dy}\right)$$

$$\frac{1}{\mu}\frac{\partial p}{\partial x} = \frac{\partial}{\partial y}\left(\frac{\partial u}{\partial y}\right)$$

Integrating $\frac{\partial u}{\partial y} = \frac{1}{\mu}\frac{\partial p}{\partial x} y + C_1$

$$\boxed{u = \frac{1}{\mu}\frac{\partial p}{\partial x}\frac{y^2}{2} + C_1 y + C_2}$$

Boundary Conditions. $\left\{ \begin{array}{l} \text{at } y=0 \\ u=0 \end{array} \right\}$ $\left\{ \begin{array}{l} \text{at } y=B \\ u=0 \end{array} \right\}$

$$\boxed{u = \frac{1}{2\mu}\left(-\frac{dp}{dx}\right)(By - y^2)}$$

Parabolic

u_{max} at $y = \frac{B}{2}$

$$u_{max} = \frac{B^2}{8\mu}\left(-\frac{dp}{dx}\right)$$

$$\cancel{\boxed{u_{mean} = \frac{2}{3} u_{max}}}$$

$$\cancel{\boxed{\frac{3}{2} u_{mean} = \frac{B^2}{8\mu}\left(-\frac{dp}{dx}\right)}}$$

$$\cancel{u_{mean} = \frac{B^2}{12\mu}\left(-\frac{dp}{dx}\right)}$$

Laminar flow between parallel plates (Cont.)

$$\boxed{u = \frac{1}{2\mu}\left(-\frac{\partial p}{\partial x}\right)(By - y^2)}$$ Parabolic velocity distribution

$u = u_{max}$ at $y = B/2$

$$u_{max} = \frac{1}{2\mu}\left(-\frac{\partial p}{\partial x}\right)\left(\frac{B^2}{2} - \frac{B^2}{4}\right)$$

$$= \frac{B^2}{8\mu}\left(-\frac{\partial p}{\partial x}\right)$$

Ratio of u_{mean} to u_{max}

Discharge per unit width of plate is given as

Small discharge dq is passing through small strip dy

$$q = \int_0^B u\,dy$$

$$= \int_0^B \frac{1}{2\mu}\left(-\frac{\partial p}{\partial x}\right)(By - y^2)\,dy$$

$$= -\frac{1}{2\mu}\frac{\partial p}{\partial x} \cdot \left[\frac{By^2}{2} - \frac{y^3}{3}\right]_0^B$$

$$q = -\frac{1}{2\mu} \cdot \frac{\partial p}{\partial x} \cdot \frac{B^3}{6} = \frac{B^3}{12\mu}\left(-\frac{\partial p}{\partial x}\right)$$

$$= A \times U_{mean}$$

$$(B \times 1)$$

$$U_{mean} = \frac{B^2}{12\mu}\left(-\frac{\partial p}{\partial x}\right)$$

$$\boxed{U_{mean} = \frac{2}{3}U_{max}}$$

$$\left(-\frac{\partial p}{\partial x}\right) = \frac{12\mu U}{B^2}$$

$$\int_{P_1}^{P_2} -(\partial p) = \frac{12\mu U}{B^2}\int_{x_1}^{x_2}\partial x$$

$$\frac{P_1 - P_2}{L} = \frac{12\mu U}{B^2}$$

$$\boxed{P_1 - P_2 = \frac{12\mu V L}{B^2}}$$

$$h_f = \frac{P_1 - P_2}{\rho g}$$

$$\boxed{h_f = \frac{12\mu V L}{\rho g B^2}}$$

Shear Stress

$$\tau = \mu \frac{\partial u}{\partial y}$$

$$= \mu \frac{\partial}{\partial y}\left[\frac{1}{2\mu}\left(-\frac{\partial p}{\partial x}\right)(By - y^2)\right]$$

$$= \left(-\frac{\partial p}{\partial x}\right)\frac{1}{2}\left[\frac{B}{2} - \frac{2y}{2}\right]$$

$$\boxed{\tau = -\frac{\partial p}{\partial x}\left[\frac{B}{2} - y\right]}$$

Shear stress varies linearly with the distance from boundary.

at $y = 0$, and $y = B$ $\quad \tau = \tau_0$ [maxm. shear stress]

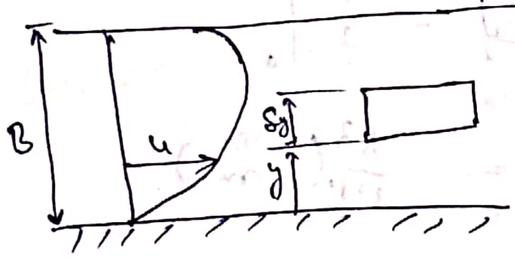
$$\tau_0 = -\frac{\partial p}{\partial x}\left(\frac{B}{2}\right)$$

at $y = B/2 \quad = \quad \boxed{\tau = 0}$

Laminar flow between parallel plates: one plate moving and other rest

Couette flow

moving plate $\longrightarrow V$



fixed plate

$$\boxed{u = \frac{1}{\mu}\frac{\partial p}{\partial x}\frac{y^2}{2} + C_1 y + C_2}$$

$u = 0$ at $y = 0$, $\quad u = V$ at $y = B$

$C_2 = 0$, $\quad \frac{V}{B} - \frac{1}{\mu}\frac{\partial p}{\partial x}\frac{B}{2} = C_1$

$$\boxed{u = \frac{V}{B}y - \frac{1}{2\mu}\left(\frac{\partial p}{\partial x}\right)\left[By - y^2\right]}$$ Parabolic

locating maxm. velocity

$$\frac{du}{dy} = 0 \Rightarrow \frac{V}{B} - \frac{1}{2\mu}\left(\frac{\partial p}{\partial x}\right)\left[B - 2y\right] = 0$$

$$\frac{V}{B} = \frac{1}{2\mu}\frac{\partial p}{\partial x}\left[B - 2y\right]$$

$$\frac{2\mu \frac{V}{B}}{\partial p/\partial x} = B - 2y \Rightarrow \boxed{y_{max} = \frac{B}{2} - \frac{\mu V}{B\, \partial p/\partial x}}$$

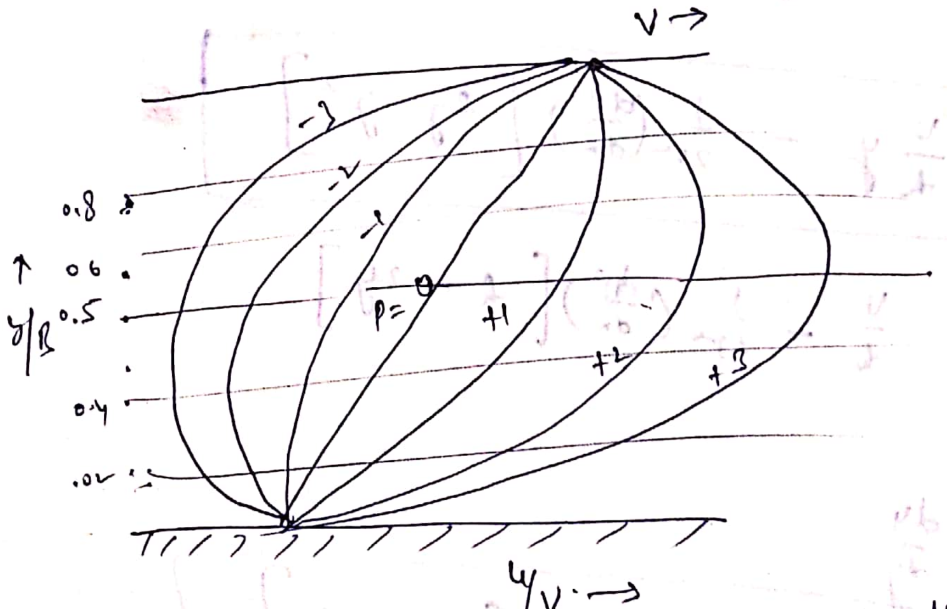
$$u_{max} = \frac{V}{B}\left(\frac{B}{2} - \frac{\mu V}{B\frac{\partial p}{\partial x}}\right) - \frac{1}{2\mu}\frac{\partial p}{\partial x}\left[B\cdot\frac{B}{2} - \frac{\mu V}{\frac{\partial p}{\partial x}} - \left(\frac{B^2}{4} + \frac{\mu^2 V^2}{B^2\left(\frac{\partial p}{\partial x}\right)^2} - \frac{B\cdot\mu V}{B\,\frac{\partial p}{\partial x}}\right)\right]$$

$$\boxed{u_{max} = \frac{V}{2} - \frac{B^2}{8\mu}\frac{\partial p}{\partial x}}$$

$$\boxed{\frac{u}{V} = \frac{y}{B} + \frac{B^2}{2\mu V}\left(-\frac{\partial p}{\partial x}\right)\frac{y}{B}\left[1 - y/B\right]}$$

$$\boxed{\frac{u}{V} = y/B + P\left[1 - y/B\right] y/B}$$

$$P = \frac{B^2}{2\mu V}\left(-\frac{\partial p}{\partial x}\right)$$ = dimensionless pressure gradient.



Non-dimensional velocity distribution Curve for Couette flow.

a family of velocity distribution curve can be plotted in terms of u/V & y/B

+ve P ⇒ Pressure drop in direction of flow

−ve P ⇒ increase in fluid pressure in direction of flow.

P = 0 ⇒ No pressure gradient (Called simple shear flow)

10 Que oil flows between two parallel plates one of which is at rest and other is moving with a velocity V. The velocity distribution between them is given by

$$u = \frac{1}{2\mu}\left(\frac{dp}{dx}\right)y^2 + C_1 y + C_2 \, , \quad \text{where } C_1 \text{ and } C_2$$

being constants of integration. Find the relation between V and $\frac{dp}{dx}$ for the following cases

(i) wall shearing stress at fixed plate is zero.
(ii) wall shearing stress at moving plate is zero.

Sol^n: $u = \frac{1}{2\mu}\left(\frac{dP}{dx}\right)y^2 + C_1 y + C_2$ B.C. $u = 0$ at $y = 0$
$u = V$ $y = h$

$C_2 = 0$,

$$V = \frac{1}{2\mu}\left(\frac{dP}{dx}\right)h^2 + C_1 h$$

$$C_1 = \frac{V}{h} + \frac{1}{2\mu}\left(-\frac{dP}{dx}\right)h$$

$$u = \frac{1}{2\mu}\left(\frac{dP}{dx}\right)y^2 + \left[\frac{V}{h} + \frac{1}{2\mu}\left(\frac{-dP}{dx}\right)h\right]y$$

$$\boxed{u = \frac{V}{h}y - \frac{1}{2\mu}\left(\frac{dP}{dx}\right)\left[hy - y^2\right]}$$

$$\frac{du}{dy} = \frac{V}{h} - \frac{1}{2\mu}\left(\frac{dP}{dx}\right)\left[h - 2y\right]$$

$$\tau = \mu\frac{du}{dy}$$

$$\boxed{\tau = \left[\frac{\mu V}{h} - \frac{1}{2}\left(\frac{dP}{dx}\right)\left\{h - 2y\right\}\right]}$$

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(i) $\tau = 0$ at $y = 0$

$$0 = \frac{\mu V}{h} - \frac{1}{2}\frac{dp}{dx}\left\{h\right\}$$

$$\Rightarrow \boxed{\frac{dp}{dx} = \frac{2\mu V}{h^2}}$$

(ii) $\tau = 0$ at $y = h$

$$0 = \left[\frac{\mu V}{h} - \frac{1}{2}\left(\frac{dp}{dx}\right)\left\{h - 2h\right\}\right]$$

$$\frac{\mu V}{h} = -\frac{1}{2}\frac{dp}{dx}h$$

$$\boxed{\frac{dp}{dx} = \frac{-2\mu V}{h^2}}$$

Tutorial problem (Solved)

An oil having viscosity of 0.0098 kgf-second/m^2 and a specific gravity of 1.59 flows through a horizontal pipe of 5 cm diameter with a pressure drop of 0.06 kgf/cm^2 per meter length of pipe.

Determine (i) flow rate in kg/min.

(ii) Shear stress at the pipe wall.

(iii) The total drag force for 100 m length of pipe, and

(iv) The power required for the 100 m length of pipe to maintain the flow.

Solution:

$$\mu = .0098 \ kgf\,second/m^2 = .0098 \times 9.8 \ \frac{N\text{-}s}{m^2} = .096 \ \frac{Nm}{m^2}$$

pressure drop = .06 kgf/cm^2 per m pipe length

$$\Delta P = \frac{.06 \times 9.8}{10^{-4}} \ N/m^2 \text{ per m pipe length} = 5880$$

$$\frac{dp}{dx} = -5880 \ N/m^2 \text{ per meter pipe length}$$

$$Q = \frac{-\pi D^4}{128\mu} \frac{dp}{dx}$$

(i) $$Q = \frac{-3.14 \times (.05)^4}{128 \times .096} \times -5880$$

$$= .00939 \ m^3/s$$

$$\rho = 1.59 \times 1000 = 1590 \ kg/m^3$$

$$\dot{m} = \rho Q = 1590 \times .00939 = 14.93 \ kg/s = 895.80 \ kg/min.$$

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(ii) Shear Stress at pipe wall $= \tau_{max.}$

$$\tau_{max} = -\frac{dp}{dx}\frac{R}{2}$$

$$= -(-5880) \times \frac{.05/2}{2} = 73.5\ N/m^2$$

(iii)

Total drag force on 100 m length of pipe

$$= \tau_{max} \times 2\pi R \times l$$

$$= 73.5 \times 2 \times 3.14 \times \frac{.05}{2} \times 100$$

$$= 1153.95\ N$$

$$= 1.154\ kN$$

(iv) Power required to maintain the flow

$$= (P_1 - P_2)\ \pi R^2\ U_{mean}$$

$$U_{mean} = \frac{U_{max.}}{2}$$

$$U_{max.} = -\frac{1}{4\mu}\frac{dp}{dx}R^2$$

$$= -\frac{1}{4 \times .096} \times (-5880) \times (.025)^2$$

$$= 9.57\ m/s$$

$$U_{mean} = 4.785\ m/s$$

$$\text{Power} = 5880 \times 3.14 \times (.025)^2 \times 4.785$$

$$= 55.216\ W/\text{metre length}$$

for 100 m length $= 5521\ W$

$$= 5.521\ kW.$$